SIT720

Task 2.1P
Data Wrangling

Lesson Review and Problem-Solving Report

Student name	Zenith Anthony Dabarera
Student ID	s225534396
Unit	SIT720 Machine Learning
Task	2.1P - Data Wrangling
Date	13 July 2026

Submission package: PDF report + Jupyter notebook

1. Task Overview

Task 2.1P requires two items: a PDF lesson-review report for Weeks 1 and 2, and a Jupyter notebook that completes seven data-wrangling questions using the City of Melbourne microclimate sensor dataset. The report also requires evidence of completing quizzes 1.28 and 2.14 with a score above 85%.

Important before submission
Run the accompanying notebook from the first cell to the last, save it with all outputs, copy the automatically printed answers for the noisiest area and windiest 2025 month into this report, and insert your own quiz screenshots. The dataset is updated over time, so the exact numerical output should come from the file used for your submission.

2. Week 1 Learning Summary

Week 1 introduced the role of data and computation in machine learning. A machine-learning workflow begins with a clear problem, relevant data and a defined outcome. Before any model is trained, the analyst must understand what each row and column represents, inspect the data types and check whether the dataset is suitable for the intended purpose.

- Python and Jupyter notebooks provide an interactive environment in which code, outputs, graphs and explanations can be kept together.
- Variables, strings, lists, dictionaries, conditions, loops and functions provide the programming foundations required to manipulate data.
- NumPy supports fast numerical arrays and vectorised operations, while pandas provides labelled Series and DataFrame structures.
- Reading a dataset is followed by checking its shape, column names, data types, sample rows and descriptive statistics.
- Reproducibility requires clear code, documented assumptions, saved outputs and a consistent sequence of processing steps.

3. Week 2 Learning Summary

Week 2 focused on data wrangling and preprocessing. Real datasets frequently contain missing entries, inconsistent data types, combined variables, extreme values and features measured on very different scales. These issues need to be addressed before reliable analysis or machine learning can occur.

- Missing-data analysis identifies which features are incomplete and how serious the incompleteness is.
- Mean imputation is simple but sensitive to outliers. Median imputation is more robust for skewed numerical variables.
- Categorical and structured text variables require transformations that preserve their meaning rather than arbitrary numerical labels.
- Histograms reveal distribution shape, concentration, skewness and unusual values. Scatter plots and correlation coefficients help describe relationships between variables.

- Min-max scaling maps values to a common range, usually 0 to 1. This prevents a feature with large units from dominating algorithms that depend on distances or numerical magnitude.
- Preprocessing choices should be explained and connected to the properties of the data, rather than applied automatically.

4. Reading List

Resource	Topic	Main learning
City of Melbourne Open Data Portal	Microclimate sensor dataset	Understand the source, update frequency, fields and possibility of missing readings.
pandas: read_csv and DataFrame	Data loading and inspection	Load tabular data, inspect columns and use labelled operations.
pandas: Working with missing data	Missing-value detection and replacement	Use isna, fillna, aggregation and carefully selected imputation.
Matplotlib histogram documentation	Distribution visualisation	Choose bins, labels and views that reveal skewness and outliers.
scikit-learn MinMaxScaler	Feature scaling	Transform each feature to a specified range while preserving relative order.

5. Reflection on Learning

The most important lesson from these two weeks is that machine learning quality depends heavily on data preparation. Before this task, it was easy to think that the main work begins when an algorithm is selected. The microclimate dataset shows that meaningful analysis begins much earlier: column names must be understood, missing readings must be counted, data types must be corrected and every transformation must have a reason.

I also learned that there is rarely one preprocessing method that is correct in every situation. For example, the mean may be reasonable for a roughly symmetric feature, but the median is safer for particulate-matter readings with extreme spikes. Similarly, latitude and longitude should not be treated as category numbers because they contain continuous geographic information. The choice of method affects the meaning of the processed data and therefore affects any model that uses it later.

The task also improved my ability to combine programming with statistical interpretation. A graph is not complete without an explanation, and a correlation coefficient should not be described as proof of causation. In future machine-learning tasks, I will inspect the data and document preprocessing decisions before training a model. This will make my work more reliable, easier to reproduce and easier to explain.

6. Problem-Solving Task

6.1 Dataset and initial audit

The notebook loads the City of Melbourne Microclimate Sensors Data and standardises the 16 expected features. The dataset includes sensor identity, timestamp, location, coordinates, wind measurements, temperature, humidity, atmospheric pressure, particulate matter and noise. The first analysis prints the missing count and missing percentage for every feature.

> **Live-data note**
> The official portal is updated regularly and may not contain every expected 15-minute reading. Therefore, row counts and missing-value totals can change. The submitted notebook output is the authoritative result for the date on which it is run.

6.2 Missing-value treatment

The numerical sensor variables are filled using their observed median. This choice is suitable because environmental readings may be right-skewed and can include extreme spikes. The median is resistant to those extremes and does not move as strongly as the mean. The notebook prints a table containing each numerical feature, its number of missing values and the exact replacement median.

SensorLocation and LatLong are device metadata rather than numerical measurements. Their missing values are filled using the most frequent observed value for the same Device_id, followed by an overall mode if a device-level value is unavailable. This approach is more meaningful than assigning the same city-wide location to every missing record.

6.3 PM25 and PM10 distributions and correlation

The notebook plots full histograms and 99th-percentile views for PM25 and PM10. The expected pattern is a concentration of low values with a long right tail and a smaller number of large readings. This is indicated when the mean is above the median, skewness is positive and the maximum is much larger than the 99th percentile.

Pearson and Spearman correlations are both reported. A positive result means that higher PM25 readings tend to occur with higher PM10 readings. Pearson measures linear association and is more sensitive to extreme values. Spearman is rank-based and is useful when the variables are skewed or their relationship is monotonic but not perfectly linear. Correlation describes association and does not by itself prove causation.

6.4 Encoding LatLong

The LatLong string is split at the comma into Latitude and Longitude, and both new columns are converted to floating-point numbers. This is direct numerical encoding of the two continuous coordinates. It preserves their order and geographic meaning. Label encoding would create arbitrary category numbers, while one-hot encoding would create unnecessary sparse columns and would not preserve distance.

6.5 Min-max scaling

Min-max scaling is applied to the continuous sensor features and the two coordinate features. Each selected feature is transformed so that its minimum becomes 0 and its maximum becomes 1. The before-and-after histograms retain the same relative distribution shape because the transformation is linear. The visible change is the x-axis range. Features with extreme outliers may show most ordinary observations compressed into a narrow part of the 0-1 range.

Wind direction is numerically continuous but conceptually circular. For a later predictive model, sine and cosine encoding would usually represent it better because 359 degrees and 0 degrees are close. The notebook still applies min-max scaling because this is explicitly requested by the task.

6.6 Area with the biggest noise

The notebook groups the original observed Noise values by SensorLocation and calculates the mean, median, maximum and number of valid observations. The area with the highest mean is selected as the noisiest area. Mean noise is used for the main answer because a single maximum may be a temporary spike; the median and count are included to check whether the result is supported by many readings.

Exact result from the notebook
After running the notebook, replace this sentence with: "[AREA] had the highest mean observed noise of [VALUE], based on [COUNT] valid readings. Its median was [VALUE] and its maximum was [VALUE]."

6.7 Windiest month in 2025

The timestamp is parsed and converted to Melbourne local time. Records from 2025 are grouped by calendar month using AverageWindSpeed. The month with the highest mean is selected. Mean wind speed is more appropriate than the sum because months may contain different numbers of valid observations. The notebook also reports the median, maximum and count for each month.

Exact result from the notebook
After running the notebook, replace this sentence with: "[MONTH] had the highest mean AverageWindSpeed in 2025 at [VALUE], based on [COUNT] valid readings."

7. Notebook Design and Reproducibility

The accompanying notebook is organised in the same order as the seven task questions. It includes explanations before each code section and automatically prints a final result summary. The code supports both a local CSV file and the official online export, handles alternative portal column labels, converts numerical columns safely and uses the original observed values for the location and month comparisons.

Step	Evidence generated
Load and standardise	Dataset source, shape, first rows, columns and data types
Audit missing data	Missing count and percentage by feature
Impute	Replacement-value table and remaining-missing check
Explore PM variables	Histograms, diagnostics, scatter plot and two correlations
Encode and scale	Latitude/Longitude values, range table and before/after plots
Answer Q6 and Q7	Ranked noise table, monthly wind table, graphs and final printed answers

8. Quiz Evidence

The task requires screenshots of the Week 1 and Week 2 quiz results. These screenshots must be taken from the student account and cannot be generated as part of the code or report.

INSERT QUIZ 1.28 SCREENSHOT HERE

The screenshot must show your own score above 85%.

INSERT QUIZ 2.14 SCREENSHOT HERE

The screenshot must show your own score above 85%.

9. Conclusion

This task demonstrates a complete introductory data-wrangling workflow. The dataset is loaded and inspected, missing values are treated with justified methods, distributions and correlations are analysed, geographic information is encoded, continuous features are scaled and grouped summaries are used to answer practical questions. The main conclusion is that preprocessing is not a mechanical step: each decision should preserve the meaning of the data and be supported by evidence.

References

City of Melbourne. (n.d.). Microclimate sensors data. City of Melbourne Open Data Portal. https://data.melbourne.vic.gov.au/explore/dataset/microclimate-sensors-data/

pandas development team. (2026). pandas documentation: Working with missing data. https://pandas.pydata.org/docs/user_guide/missing_data.html

pandas development team. (2026). pandas.read_csv documentation. https://pandas.pydata.org/docs/reference/api/pandas.read_csv.html

Matplotlib development team. (2026). Histograms. https://matplotlib.org/stable/gallery/statistics/hist.html

scikit-learn developers. (2026). MinMaxScaler. https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.MinMaxScaler.html